

QUOC ANH TRINH (ANDREW)

Data Scientist — M.Sc in Business Analytics, Deakin University



📞 : +61 451 886 468 — ✉ : andrews.trinh@gmail.com — 🌐 : qatrinh

DATA SCIENTIST

- Purpose-driven Data Scientist with 5+ years of experience delivering end-to-end AI/ML solutions in FinTech and banking. Expertise in leveraging Generative AI, Large Language Models (LLMs), and traditional predictive modelling to drive data-led decision-making. Proven track record in automating complex regulatory workflows, optimizing transaction monitoring for institutions like Bendigo & Adelaide Bank, and building scalable data products that translate technical complexity into measurable social and business impact.

DATA SCIENCE & AI EXPERTISE

AI Innovation: (LLMs, NLP, Python)	• Developed agentic AI architectures to extract and transform 156 AUSTRAC docs into 198 actionable typologies and 300+ indicators • Co-designed automated UAR-to-SMR pipeline using LLMs and data masking, reducing regulatory report prep time by 80% • Fine-tuned predictive scenarios for high-stakes risk detection achieving 60-70% true positive rates
Machine Learning: (Scikit-learn, Spark)	• Calibrated transaction monitoring models using genetic algorithms and statistical trend modelling, achieving 67% alert reduction • Automated marketing via customer classification models driving 30% CTR uplift
Infrastructure & MLOps: (AWS, GCP, Git, Docker)	• Optimized ETL pipelines in Redshift and AWS Glue, reducing cloud costs by 30% and dashboard runtime 70% • Engineered automated workflows processing 150+ alerts daily with 99.9% integrity using CI/CD principles
Analytics & Insight: (SQL, PowerBI, R)	• Engineered Power BI dashboards eliminating 70% of ad-hoc requests, reducing prioritization time by 60% • Tutored 100+ students in SQL and Python, specializing in translating technical findings for non-technical audiences

PROFESSIONAL EXPERIENCE

Analyst — Lead of Intelligence & Analytics <i>Financial Crime Risk - Intelligence and Analytics</i>	Bendigo & Adelaide Bank	05/2021 - Present
• End-to-End AI Solutions: Led the transformation of analytical capabilities using machine learning and NLP for exploratory analysis and predictive SMR-triggered alert sets. • Generative AI Innovation: Engineered an AI-driven typology management platform, applying advanced ETL to transform 156 national regulatory documents into actionable data assets. • ML Modelling: Built and fine-tuned transaction monitoring scenarios utilizing statistical confidence testing and statistical recalibration to save 300+ investigation hours annually. • Strategic Oversight: Advised transformation and governance teams on dashboard design for alert prioritization and operational KPIs, enhancing data-led decision-making.		

KEY DATA SCIENCE ACHIEVEMENTS

- **AI reporting pipeline:** Co-designed the regulated UAR-to-SMR pipeline using Selenium, LLMs, and SQL data modelling, **automating report prep by 80%**.
- **Algorithm fine-tuning:** Leveraged genetic algorithms to achieve **60-70% true positive rates** in critical risk scenarios while reducing alert volume by **67%**.
- **Data Asset Transformation:** Transformed AUSTRAC documents into **198 typologies** and **300+ indicators** using agentic AI, forming the baseline for modern, data-led monitoring.

- **Scalable Architecture:** Engineered SQL-based replication of **40+ scenarios** for a major bank migration, achieving **100% scenario alignment**.

Data Analyst

Remitano

03/2020 - 04/2021

International cryptocurrency exchange operating in 30+ countries.

- **Predictive Analytics:** Built and automated marketing predictive models boosting click-through rates by **30%**.
- **ETL Optimization:** Slashed cloud costs by **30%** and runtime by **70%** through optimized ETL pipelines and data normalization in Redshift.
- **Complex Modelling:** Engineered recursive SQL for MLM commission calculations on **150K+ relationships**, eliminating high-cost graph database dependencies.

EDUCATION

- **Master of Business Analytics** – Deakin University, 2019
- **Bachelor of General Business Management** – National Economics University, 2013

HONORS & AWARDS

- Third place & Honorable Mention - Project SafeAus - GovHack Australia, 09/2021
- #13 in public test and #32 in private test - Pet finder Kaggle Competition, 2019
- Won 2 Open Gov hackathon prizes for impactful data-driven solutions

VOLUNTEER ACTIVITIES & HOBBIES

- Tutored 100+ students in SQL and Python, bridging technical and non-technical communication gaps.
- Hobbies: Sports (Football), Building data-driven solutions for micro-businesses.